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References

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Adolf et al.: Optimal sampling rates for reliable continuous-time first-order autoregressive and vector autoregressive modeling Adolf-Loossens-Tuerlinckx-etal-2021

Janne K. Adolf et al. "Optimal sampling rates for reliable continuous-time first-order autoregressive and vector autoregressive modeling". In: *Psychological Methods* 26.6 (Dec. 2021), pp. 701–718. ISSN: 1082-989X. DOI: 10.1037/met0000398.

Abstract: Autoregressive and vector autoregressive models are a driving force in current psychological research. In affect research they are, for instance, frequently used to formalize affective processes and estimate affective dynamics. Discrete-time model variants are most commonly used, but continuous-time formulations are gaining popularity, because they can handle data from longitudinal studies in which the sampling rate varies within the study period, and yield results that can be compared across data sets from studies with different sampling rates. However, whether and how the sampling rate affects the quality with which such continuous-time models can be estimated, has largely been ignored in the literature. In the present article, we show how the sampling rate affects the estimation reliability (i.e., the standard errors of the parameter estimators, with smaller values indicating higher reliability) of continuous-time autoregressive and vector autoregressive models. Moreover, we determine which sampling rates are optimal in the sense that they lead to standard errors of minimal size (subject to the assumption that the models are correct). Our results are based on the theories of optimal design and maximum likelihood estimation. We illustrate them making use of data from the COGITO Study. We formulate recommendations for study planning, and elaborate on strengths and limitations of our approach.

Ash et al.: Sensitivity, specificity, and tolerability of the BACTrack Skyn compared to other alcohol monitoring approaches among young adults in a field-based setting
Ash-Gueorguieva-Barnett-etal-2022

Garrett I. Ash et al. "Sensitivity, specificity, and tolerability of the BACTrack Skyn compared to other alcohol monitoring approaches among young adults in a field-based setting". In: *Alcoholism: Clinical and Experimental Research* 46.5 (May 2022), pp. 783–796. ISSN: 1530-0277. DOI: 10.1111/acer.14804.

Abstract: Background: There is a need for novel alcohol biosensors that are accurate, able to detect alcohol concentration close in time to consumption, and feasible and acceptable for many clinical and research applications. We evaluated the field accuracy and tolerability of novel (BACTrack Skyn) and established (Alcohol Monitoring Systems SCRAM CAM) alcohol biosensors. Methods: The sensor and diary data were collected in a larger study of a biofeedback intervention and compared observationally in the present sub-study. Participants (high-risk drinkers, 40% female; median age 21) wore both Skyn and SCRAM CAM sensors for 1-6 days and were instructed to drink as usual. Data from the first cohort of participants ($N = 27$; 101 person-days) were used to find threshold values of transdermal alcohol that classified each day as meeting or not meeting defined levels of drinking (heavy, above-moderate, any). These values were used to develop scoring metrics that were subsequently tested using the second cohort ($N = 20$; 57 person-days). Data from both biosensors were compared to mobile diary self-report to evaluate sensitivity and specificity in relation to a priori standards established in the literature. Results: Skyn classification rules for Cohort #1 within 3 months of device shipment showed excellent sensitivity for heavy drinking (94%) and exceeded expectations for above-moderate and any drinking (78% and 69%, respectively), while specificity met expectations (91%). However, classification worsened when Cohort #1 devices ≥ 3 months from shipment were tested (area under curve for receiver operator characteristic 0.87 vs. 0.79) and the derived classification threshold when applied to Cohort #2 was inadequately specific (70%). Skyn tolerability metrics were excellent and exceeded the SCRAM CAM ($p \leq 0.001$). Con-

clusions: Skyn tolerability was favorable and accuracy rules were internally derivable but did not yield useful scoring metrics going forward across device lots and months of usage.

Baek et al.: Meta-analysis of single-case experimental design using multilevel modeling

Baek-Luo-Lam-2023

Eunkyeng Baek, Wen Luo, and Kwok Hap Lam. "Meta-analysis of single-case experimental design using multilevel modeling". In: *Behavior Modification* 47.6 (Jan. 2023), pp. 1546–1573. ISSN: 1552-4167. DOI: 10.1177/01454455221144034.

Abstract: Multilevel modeling (MLM) is an approach for meta-analyzing single-case experimental designs (SCED). In this paper, we provide a step-by-step guideline for using the MLM to meta-analyze SCED time-series data. The MLM approach is first presented using a basic three-level model, then gradually extended to represent more realistic situations of SCED data, such as modeling a time variable, moderators representing different design types and multiple outcomes, and heterogeneous within-case variance. The presented approach is then illustrated using real SCED data. Practical recommendations using the MLM approach are also provided for applied researchers based on the current methodological literature. Available free and commercial software programs to meta-analyze SCED data are also introduced, along with several hands-on software codes for applied researchers to implement their own studies. Potential advantages and limitations of using the MLM approach to meta-analyzing SCED are discussed.

Bakk et al.: Relating latent class membership to external variables: An overview

Bakk-Kuha-2020

Zsuzsa Bakk and Jouni Kuha. "Relating latent class membership to external variables: An overview". In: *British Journal of Mathematical and Statistical Psychology* 74.2 (Nov. 2020), pp. 340–362. ISSN: 2044-8317. DOI: 10.1111/bmsp.12227.

Abstract: In this article we provide an overview of existing approaches for relating latent class membership to external variables of interest. We extend on the work of Nylund-Gibson et al. (Structural Equation Modeling: A Multidisciplinary Journal, 2019, 26, 967), who summarize models with distal outcomes by providing an overview of most recommended modeling options for models with covariates and larger models with multiple latent variables as well. We exemplify the modeling approaches using data from the General Social Survey for a model with a distal outcome where underlying model assumptions are violated, and a model with multiple latent variables. We discuss software availability and provide example syntax for the real data examples in Latent GOLD.

L. F. Bringmann: Person-specific networks in psychopathology: Past, present, and future **Bringmann-2021**

Laura F. Bringmann. "Person-specific networks in psychopathology: Past, present, and future". In: *Current Opinion in Psychology* 41 (Oct. 2021), pp. 59–64. ISSN: 2352-250X. DOI: 10.1016/j.copsyc.2021.03.004.

Abstract: In the psychological network approach, mental disorders such as major depressive disorder are conceptualized as networks. The network approach focuses on the symptom structure or the connections between symptoms instead of the severity (i.e., mean level) of a symptom. To infer a person-specific network for a patient, time-series data are needed. By far the most common model to statistically model the person-specific interactions between symptoms or momentary states has been the vector autoregressive (VAR) model. Although the VAR model helps to bring psychological network theory into clinical research and closer to clinical practice, several discrepancies arise when we map the psychological network theory onto the VAR-based network models. These challenges and possible solutions are discussed in this review.

Man Chen and James E. Pustejovsky. "Multilevel meta-analysis of single-case experimental designs using robust variance estimation." In: *Psychological Methods* 29.3 (June 2024), pp. 537–560. ISSN: 1082-989X. DOI: 10.1037/met0000510.

Abstract: Single-case experimental designs (SCEDs) are used to study the effects of interventions on the behavior of individual cases, by making comparisons between repeated measurements of an outcome under different conditions. In research areas where SCEDs are prevalent, there is a need for methods to synthesize results across multiple studies. One approach to synthesis uses a multilevel meta-analysis (MLMA) model to describe the distribution of effect sizes across studies and across cases within studies. However, MLMA relies on having accurate sampling variances of effect size estimates for each case, which may not be possible due to auto-correlation in the raw data series. One possible solution is to combine MLMA with robust variance estimation (RVE), which provides valid assessments of uncertainty even if the sampling variances of effect size estimates are inaccurate. Another possible solution is to forgo MLMA and use simpler, ordinary least squares (OLS) methods with RVE. This study evaluates the performance of effect size estimators and methods of synthesizing SCEDs in the presence of auto-correlation, for several different effect size metrics, via a Monte Carlo simulation designed to emulate the features of real data series. Results demonstrate that the MLMA model with RVE performs properly in terms of bias, accuracy, and confidence interval coverage for estimating overall average log response ratios. The OLS estimator corrected with RVE performs the best in estimating overall average Tau effect sizes. None of the available methods perform adequately for meta-analysis of within-case standardized mean differences.

M. W.-L. Cheung: Synthesizing indirect effects in mediation models with meta-analytic methods **Cheung-2021**

Mike W.-L. Cheung. "Synthesizing indirect effects in mediation models with meta-analytic methods". In: *Alcohol and Alcoholism* 57.1 (June 2021), pp. 5–15. DOI: 10.1093/alcalc/agab044.

Abstract: Aims: A mediator is a variable that explains the underlying mechanism between an independent variable and a dependent variable. The indirect effect indicates the effect from the predictor to the outcome variable via the mediator. In contrast, the direct effect represents the predictor's effort on the outcome variable after controlling for the mediator. Methods: A single study rarely provides enough evidence to answer research questions in a particular domain. Replications are generally recommended as the gold standard to conduct scientific research. When a sufficient number of studies have been conducted addressing similar research questions, a meta-analysis can be used to synthesize those studies' findings. Results: The main objective of this paper is to introduce two frameworks to integrating studies using mediation analysis. The first framework involves calculating standardized indirect effects and direct effects and conducting a multivariate meta-analysis on those effect sizes. The second one uses meta-analytic structural equation modeling to synthesize correlation matrices and fit mediation models on the average correlation matrix. We illustrate these procedures on a real dataset using the R statistical platform. Conclusion: This paper closes with some further directions for future studies.

S. F. Cheung et al.: Improving an old way to measure moderation effect in standardized units. **Cheung-Cheung-Lau-etal-2022**

Shu Fai Cheung, Sing-Hang Cheung, et al. "Improving an old way to measure moderation effect in standardized units." In: *Health Psychology* 41.7 (July 2022), pp. 502–505. ISSN: 0278-6133. DOI: 10.1037/hea0001188.

Abstract: Moderation effects in multiple regression, tested usually by the inclusion of a product term, are frequently investigated in health psychology. However, several issues in presenting the moderation effects in standardized units and their associated confidence intervals are commonly observed. While an old method had been proposed to standardize variables in moderated regression before fitting a moderated regression model, this method was rarely used due to inconvenience and even when used, the confidence intervals derived were biased. Here, we attempt to solve these two problems by providing a tool to conveniently conduct standardization in moderated regression without the step of standardizing the variables beforehand and to accurately form the nonparametric bootstrapping confidence intervals for this standardized measure of moderation effects. Health psychology researchers are now equipped with a tool that can be used to report and interpret standardized moderation effects correctly.

S. F. Cheung et al.: FINDOUT: Using either SPSS commands or graphical user interface to identify influential cases in structural equation modeling in AMOS

Cheung-Pesigan-2023a

Shu Fai Cheung and Ivan Jacob Agaloos Pesigan. "FINDOUT: Using either SPSS commands or graphical user interface to identify influential cases in structural equation modeling in AMOS". In: *Multivariate Behavioral Research* 58.5 (Jan. 2023), pp. 964–968. DOI: 10.1080/00273171.2022.2148089.

Abstract: The results in a structural equation modeling (SEM) analysis can be influenced by just a few observations, called influential cases. Tools have been developed for users of R to identify them. However, similar tools are not available for AMOS, which is also a popular SEM software package. We introduce the FINDOUT toolset, a group of SPSS extension commands, and an AMOS plugin, to identify influential cases and examine how these cases influence the results. The SPSS commands can be used either as syntax commands or as custom dialogs from pull-down menus, and the AMOS plugin can be run from AMOS pull-down menu. We believe these tools can help researchers to examine the robustness of their findings to influential cases.

S. F. Cheung et al.: semlbci: An R package for forming likelihood-based confidence intervals for parameter estimates, correlations, indirect effects, and other derived parameters **Cheung-Pesigan-2023b**

Shu Fai Cheung and Ivan Jacob Agaloos Pesigan. "semlbci: An R package for forming likelihood-based confidence intervals for parameter estimates, correlations, indirect effects, and other derived parameters". In: *Structural Equation Modeling: A Multidisciplinary Journal* 30.6 (May 2023), pp. 985–999. DOI: 10.1080/10705511.2023.2183860.

Abstract: There are three common types of confidence interval (CI) in structural equation modeling (SEM): Wald-type CI, bootstrapping CI, and likelihood-based CI (LBCI). LBCI has the following advantages: (1) it has better coverage probabilities and Type I error rate compared to Wald-type CI when the sample size is finite; (2) it correctly tests the null hypothesis of a parameter based on likelihood ratio chi-square difference test; (3) it is less computationally intensive than bootstrapping CI; and (4) it is invariant to transformations. However, LBCI is not available in many popular SEM software packages. We developed an R package, semlbci, for forming LBCI for parameters in models fitted by lavaan, a popular open-source SEM package, such that researchers have more options in forming CIs for parameters in SEM. The package supports both unstandardized and standardized estimates, derived parameters such as indirect effect, multisample models, and the robust LBCI proposed by Falk.

S. F. Cheung et al.: DIY bootstrapping: Getting the nonparametric bootstrap confidence interval in SPSS for any statistics or function of statistics (when this bootstrapping is appropriate) **Cheung-Pesigan-Vong-2023**

Shu Fai Cheung, Ivan Jacob Agaloos Pesigan, and Weng Ngai Vong. "DIY bootstrapping: Getting the nonparametric bootstrap confidence interval in SPSS for any statistics or function of statistics (when this bootstrapping is appropriate)". In: *Behavior Research Methods* 55.2 (2023), pp. 474–490. DOI: 10.3758/s13428-022-01808-5.

Abstract: Researchers can generate bootstrap confidence intervals for some statistics in SPSS using the BOOTSTRAP command. However, this command can only be applied to selected procedures, and only to selected statistics in these procedures. We developed an extension command and prepared some sample syntax files based on existing approaches from the Internet to illustrate how researchers can (a) generate a large number of nonparametric bootstrap samples, (b) do desired analysis on all these samples, and (c) form the bootstrap confidence intervals for selected statistics using the OMS commands. We developed these tools to help researchers apply nonparametric bootstrapping to any statistics for which this method is appropriate, including statistics derived from other statistics, such as standardized effect size measures computed from the t test results. We also discussed how researchers can extend the tools for other statistics and scenarios they encounter.

Courtney et al.: Testing affect regulation models of drinking prior to and after drinking initiation using ecological momentary assessment **Courtney-Russell-2021**

Jimikaye B. Courtney and Michael A. Russell. "Testing affect regulation models of drinking prior to and after drinking initiation using ecological momentary assessment". In: *Psychology of Addictive Behaviors* 35.5 (Aug. 2021), pp. 597–608. ISSN: 0893-164X. DOI: 10.1037/adb0000763.

Abstract: Objective: Affect regulation models of drinking state that affect motivates and reinforces drinking. Few studies have been able to elucidate the timing of these associations in natural settings. We tested positive affect (PA) and negative affect (NA) as predictors of drinking behavior, both prior to and during drinking episodes, and whether drinking predicted changes in affect during episodes. Method: Two hundred twenty two regularly drinking young adults (21-29 years, 84% undergraduates), completed an ecological momentary assessment (EMA) protocol for five consecutive 24-hr periods stretching across 6 days (Wednesday-Monday). Participants provided PA and NA reports three times daily and every half hour during drinking episodes. Alcohol consumption reports were provided each morning and every half hour during drinking episodes. Results: Multi-level models showed that greater pre-drinking PA predicted higher odds of drinking and greater number of drinks consumed. Pre-drinking NA did not predict same day odds of drinking or drinks consumed.

Episode-level results revealed different associations for PA and NA with drinking. Current PA did not predict drinks consumed over the next half hour; however, increased drinking was associated with greater increases in PA over the next half hour. Higher NA predicted fewer drinks consumed in the next half hour and higher odds of the end of a drinking episode; however, increased drinking was not associated with changes in NA. Conclusions: PA increased following drinking during episodes. Our results suggest that a focus on PA prior to episodes and a focus on NA during episodes may interrupt processes leading to heavy drinking, and may therefore aid prevention efforts.

Declercq et al.: Multilevel Meta-Analysis of Individual Participant Data of Single-Case Experimental Designs: One-Stage versus Two-Stage Methods

Declercq-Jamshidi-FernandezCastilla-etal-2022

Lies Declercq et al. "Multilevel Meta-Analysis of Individual Participant Data of Single-Case Experimental Designs: One-Stage versus Two-Stage Methods". In: *Multivariate Behavioral Research* 57.2-3 (2022), pp. 298–317. ISSN: 1532-7906. DOI: 10.1080/00273171.2020.1822148.

Abstract: To conduct a multilevel meta-analysis of multiple single-case experimental design (SCED) studies, the individual participant data (IPD) can be analyzed in one or two stages. In the one-stage approach, a multilevel model is estimated based on the raw data. In the two-stage approach, an effect size is calculated for each participant and these effect sizes and their sampling variances are subsequently combined to estimate a meta-analytic multilevel model. The multilevel model in the two-stage approach has fewer parameters to estimate, in exchange for the reduction of information of the raw data to effect sizes. In this paper we explore how the one-stage and two-stage IPD approaches can be applied in the context of meta-analysis of single-case designs. Both approaches are compared for several single-case designs of increasing complexity. Through a simulation study we show that the two-stage approach obtains better convergence rates for more complex models, but that model estimation does not necessarily converge at a faster speed. The point estimates of the fixed effects are unbiased for both approaches across all models, as such

confirming results from methodological research on IPD meta-analysis of group-comparison designs. In light of these results, we discuss the implementation of both methods in R.

DeMartini et al.: Dynamic structural equation modeling of the relationship between alcohol habit and drinking variability DeMartini-Gueorguieva-Taylor-etal-2022

Kelly S. DeMartini et al. "Dynamic structural equation modeling of the relationship between alcohol habit and drinking variability". In: *Drug and Alcohol Dependence* 233 (Apr. 2022), p. 109202. ISSN: 0376-8716. DOI: 10.1016/j.drugalcdep.2021.109202.

Abstract: Background: A hyper-engaged habit system may be common in alcohol use disorders (AUDs). Regarding drinking patterns, habit may be expressed as higher levels of drinking autoregression, where previous day drinking is correlated with next day drinking. This study utilized dynamic structural equation models (DSEM) with intensive longitudinal data to understand whether alcohol habit relates to drinking autoregression and variable levels of alcohol consumption. Methods: Participants were adult drinkers ($N = 313$) who completed baseline self-report assessments of past 30-day alcohol consumption and alcohol habit. Alcohol habit was measured by the Self Report Habit Index (SRHI). Thirty-day coding of the Timeline Followback assessed total daily drinking and any daily heavy drinking. Results: The DSEM model for daily drinking found a weak but significant autoregressive data structure. Alcohol habit was related to increased mean drinking but did not strengthen the autoregressive effect of drinks per day. Higher alcohol habit was associated with higher levels of drinks per day person-specific variability. This pattern was replicated with the DSEM model for heavy drinking. Alcohol habit did not impact the autoregressive effect of heavy drinking but was associated with higher levels of heavy drinking. Conclusions: While both drinks per day and heavy drinking showed a significant autoregressive structure, evidence of alcohol habit did not strengthen this effect. Alcohol habit did impact drinking variability; higher alcohol habit is associated with greater levels of drinking variability and higher mean drinking. Strategies to regulate drinking variability, including heavier drinking occasions, could target AUD habit.

Didier et al.: Signal processing and machine learning with transdermal alcohol concentration to predict natural environment alcohol consumption.

Didier-King-Polley-etal-2023

Nathan A. Didier et al. “Signal processing and machine learning with transdermal alcohol concentration to predict natural environment alcohol consumption.” In: *Experimental and Clinical Psychopharmacology* 32.2 (Oct. 2023), pp. 245–254. ISSN: 1064-1297. DOI: 10.1037/pha0000683.

Abstract: Wrist-worn alcohol biosensors continuously and discreetly record transdermal alcohol concentration (TAC) and may allow alcohol researchers to monitor alcohol consumption in participants’ natural environments. However, the field lacks established methods for signal processing and detecting alcohol events using these devices. We developed software that streamlines analysis of raw data (TAC, temperature, and motion) from a wrist-worn alcohol biosensor (BACtrack Skyn) through a signal processing and machine learning pipeline: biologically implausible skin surface temperature readings (¡ 28C) were screened for potential device removal and TAC artifacts were corrected, features that describe TAC (e.g., rise duration) were calculated and used to train models (random forest and logistic regression) that predict self-reported alcohol consumption, and model performances were measured and summarized in autogenerated reports. The software was tested using 60 Skyn data sets recorded during 30 alcohol drinking episodes and 30 nonalcohol drinking episodes. Participants (N = 36; 13 with alcohol use disorder) wore the Skyn during one alcohol drinking episode and one nonalcohol drinking episode in their natural environment. In terms of distinguishing alcohol from nonalcohol drinking, correcting artifacts in the data resulted in 10% improvement in model accuracy relative to using raw data. Random forest and logistic regression models were both accurate, correctly predicting 97% (58/60; AUC-ROCs = 0.98, 0.96) of episodes. Area under TAC curve, rise duration of TAC curve, and peak TAC were the most important features for predictive accuracy. With promising model performance, this protocol will enhance the efficiency and reliability of TAC sensors for future alcohol monitoring research.

Dora et al.: The daily association between affect and alcohol use: A meta-analysis of individual participant data **Dora-Piccirillo-Foster-etal-2023**

Jonas Dora et al. "The daily association between affect and alcohol use: A meta-analysis of individual participant data". In: *Psychological Bulletin* 149.1-2 (Jan. 2023), pp. 1–24. ISSN: 0033-2909. DOI: 10.1037/bul0000387.

Abstract: Influential psychological theories hypothesize that people consume alcohol in response to the experience of both negative and positive emotions. Despite two decades of daily diary and ecological momentary assessment research, it remains unclear whether people consume more alcohol on days they experience higher negative and positive affects in everyday life. In this preregistered meta-analysis, we synthesized the evidence for these daily associations between affect and alcohol use. We included individual participant data from 69 studies (N = 12,394), which used daily and momentary surveys to assess the affect and the number of alcoholic drinks consumed. Results indicate that people are not more likely to drink on days they experience high negative affect but are more likely to drink and drink heavily on days high in positive affect. People self-reporting a motivational tendency to drink-to-cope and drink-to-enhance consumed more alcohol but not on days they experienced higher negative and positive affects. Results were robust across different operationalizations of affect, study designs, study populations, and individual characteristics. These findings challenge the long-held belief that people drink more alcohol following increase in negative affect. Integrating these findings under different theoretical models and limitations of this field of research, we collectively propose an agenda for future research to explore open questions surrounding affect and alcohol use.

Driver: Inference with cross-lagged effects—Problems in time **Driver-2025**

Charles C. Driver. "Inference with cross-lagged effects—Problems in time". In: *Psychological Methods* 30.1 (Feb. 2025), pp. 174–202. ISSN: 1082-989X. DOI: 10.1037/met0000665.

Abstract: The interpretation of cross-effects from vector autoregressive models to infer structure and causality among constructs is widespread and sometimes problematic. I describe problems in the interpretation of cross-effects when processes that are thought to fluctuate continuously in time are, as is typically done, modeled as changing only in discrete steps (as in e.g., structural equation modeling)–zeroes in a discrete-time temporal matrix do not necessarily correspond to zero effects in the underlying continuous processes, and vice versa. This has implications for the common case when the presence or absence of cross-effects is used for inference about underlying causal processes. I demonstrate these problems via simulation, and also show that when an underlying set of processes are continuous in time, even relatively few direct causal links can result in much denser temporal effect matrices in discrete-time. I demonstrate one solution to these issues, namely parameterizing the system as a stochastic differential equation and focusing inference on the continuous-time temporal effects. I follow this with some discussion of issues regarding the switch to continuous-time, specifically regularization, appropriate measurement time lag, and model order. An empirical example using intensive longitudinal data highlights some of the complexities of applying such approaches to real data, particularly with respect to model specification, examining misspecification, and parameter interpretation.

Elmer et al.: Modeling categorical time-to-event data: The example of social interaction dynamics captured with event-contingent experience sampling methods.

Elmer-vanDuijn-Ram-etal-2025

Timon Elmer et al. "Modeling categorical time-to-event data: The example of social interaction dynamics captured with event-contingent experience sampling methods." In: *Psychological Methods* 30.6 (Dec. 2025), pp. 1345–1363. ISSN: 1082-989X. DOI: 10.1037/met0000598.

Abstract: The depth of information collected in participants' daily lives with active (e.g., experience sampling surveys) and passive (e.g., smartphone sensors) ambulatory measurement methods is immense. When measuring participants' behaviors in daily life, the timing of particular events—such as social interactions—is often recorded. These data facilitate the investigation of new types of

research questions about the timing of those events, including whether individuals' affective state is associated with the rate of social interactions (binary event occurrence) and what types of social interactions are likely to occur (multicategory event occurrences, e.g., interactions with friends or family). Although survival analysis methods have been used to analyze time-to-event data in longitudinal settings for several decades, these methods have not yet been incorporated into ambulatory assessment research. This article illustrates how multilevel and multistate survival analysis methods can be used to model the social interaction dynamics captured in intensive longitudinal data, specifically *when individuals exhibit particular categories of behavior*. We provide an introduction to these models and a tutorial on how the timing and type of social interactions can be modeled using the R statistical programming language. Using event-contingent reports ($N = 150$, $N_{\text{events}} = 64,112$) obtained in an ambulatory study of interpersonal interactions, we further exemplify an empirical application case. In sum, this article demonstrates how survival models can advance the understanding of (social interaction) dynamics that unfold in daily life.

Feinn et al.: Individual differences in affect dynamics and alcohol-related outcomes

Feinn-Armeli-Tennen-2023

Richard Feinn, Stephen Armeli, and Howard Tennen. "Individual differences in affect dynamics and alcohol-related outcomes". In: *Substance Use & Misuse* 58.8 (Apr. 2023), pp. 967–974. ISSN: 1532-2491. DOI: 10.1080/10826084.2023.2201829.

Abstract: Background: To examine whether individual differences in intensive longitudinal data-derived affective dynamics (i.e. positive and negative affect variability and inertia and positive affect-negative affect bipolarity) - posited to be indicative of emotion dysregulation - are uniquely related to drinking level and affect-regulation drinking motives after controlling for mean levels of affective states. Method: We used a large sample of college student drinkers ($N = 1640$, 54% women) who reported on their affective states, drinking levels and drinking motives daily for 30 days using a web-based daily diary. We then calculated from the daily data positive and negative affect variability, inertia, affect bipolarity and mean levels of affect and used these as predictors of average drinking level and affect-regulation drinking motives (assessed using both retrospective and daily reporting methods). Results: Findings from dynamic structural equation models indicated that mean levels of affect were uniquely related to drinking motives, but not drinking level. Few dynamic affect predictors were uniquely related to outcomes in the predicted direction after controlling for mean affect levels. Conclusion: Our results add to the inconsistent literature regarding the associations between affective dynamics and alcohol-related outcomes, suggesting that any effects of these indicators, after controlling for mean affect levels, might be more complex than can be detected with simple linear models.

Fisher et al.: A square-root second-order extended Kalman filtering approach for estimating smoothly time-varying parameters Fisher-Chow-Molenaar-etal-2022

Zachary F. Fisher et al. "A square-root second-order extended Kalman filtering approach for estimating smoothly time-varying parameters". In: *Multivariate Behavioral Research* 57.1 (2022), pp. 134–152. ISSN: 1532-7906. DOI: 10.1080/00273171.2020.1815513.

Abstract: Researchers collecting intensive longitudinal data (ILD) are increasingly looking to model psychological processes, such as emotional dynamics, that organize and adapt across time in complex and meaningful ways. This is also the case for researchers looking to characterize the impact of an intervention on individual behavior. To be useful, statistical models must be capable of characterizing these processes as complex, time-dependent phenomenon, otherwise only a fraction of the system dynamics will be recovered. In this paper we introduce a Square-Root Second-Order Extended Kalman Filtering approach for estimating smoothly time-varying parameters. This approach is capable of handling dynamic factor models where the relations between variables underlying the processes of interest change in a manner that may be difficult to specify in advance. We examine the performance of our approach in a Monte Carlo simulation and show the proposed algorithm accurately recovers the unobserved states in the case of a bivariate dynamic factor model with time-varying dynamics and treatment effects. Furthermore, we illustrate

the utility of our approach in characterizing the time-varying effect of a meditation intervention on day-to-day emotional experiences.

Fridberg et al.: Examining features of transdermal alcohol biosensor readings: A promising approach with implications for research and intervention

Fridberg-Wang-Porges-2022

Daniel J. Fridberg, Yan Wang, and Eric Porges. "Examining features of transdermal alcohol biosensor readings: A promising approach with implications for research and intervention". In: *Alcoholism: Clinical and Experimental Research* 46.4 (Feb. 2022), pp. 514–516. ISSN: 1530-0277. DOI: 10.1111/acer.14794.

Georgeson et al.: A sensitivity analysis for temporal bias in cross-sectional mediation

Georgeson-AlvarezBartolo-MacKinnon-2025

A. R. Georgeson, Diana Alvarez-Bartolo, and David P. MacKinnon. "A sensitivity analysis for temporal bias in cross-sectional mediation". In: *Psychological Methods* 30.6 (Dec. 2025), pp. 1326–1344. ISSN: 1082-989X. DOI: 10.1037/met0000628.

Abstract: For over three decades, methodologists have cautioned against the use of cross-sectional mediation analyses because they yield biased parameter estimates. Yet, cross-sectional mediation models persist in practice and sometimes represent the only analytic option. We propose a sensitivity analysis procedure to encourage a more principled use of cross-sectional mediation analysis, drawing inspiration from Gollob and Reichardt (1987, 1991). The procedure is based on the two-wave longitudinal mediation model and uses phantom variables for the baseline data. After a researcher provides ranges of possible values for cross-lagged, autoregressive, and baseline Y and M correlations among the phantom and observed variables, they can use the sensitivity analysis to identify longitudinal conditions in which conclusions from a cross-sectional model would differ most from a longitudinal model. To support the procedure, we first show that differences in sign

and effect size of the b-path occur most often when the cross-sectional effect size of the b-path is small and the cross-lagged and the autoregressive correlations are equal or similar in magnitude. We then apply the procedure to cross-sectional analyses from real studies and compare the sensitivity analysis results to actual results from a longitudinal mediation analysis. While no statistical procedure can replace longitudinal data, these examples demonstrate that the sensitivity analysis can recover the effect that was actually observed in the longitudinal data if provided with the correct input information. Implications of the routine application of sensitivity analysis to temporal bias are discussed. R code for the procedure is provided in the online supplementary materials.

Gistelinck et al.: Multilevel autoregressive models for longitudinal dyadic data

Gistelinck-Loeys-2020

Fien Gistelinck and Tom Loeys. "Multilevel autoregressive models for longitudinal dyadic data". In: *TPM - Testing, Psychometrics, Methodology in Applied Psychology* 27 (2020), pp. 433–452. ISSN: 1972-6325. DOI: 10.4473/TPM27.3.7.

Abstract: In social and behavioral science, dyadic research has become more and more popular. In case of cross-sectional dyadic data, one can apply the actor-partner interdependence model (APIM). When dyads are measured repeatedly over time, applied researchers are often hesitant to analyze such data due to the statistical complexity. In this paper, we introduce a user-friendly Shiny-application, called the LDDinSEM-application. The app automatically fits the lagged dependent actor-partner interdependence model (LD-APIM), a multilevel autoregressive model extension of the APIM within the structural equation modeling (SEM) framework. The application allows the researcher to investigate the effects of an antecedent on an outcome, given the previous outcome. We illustrate the app using an empirical example assessing the actor and partner effects of positive relationship feelings on next day's intimacy in heterosexual couples.

Grayson et al.: The impact of stress on personality state expressions **Grayson-Harari-Matz-2026**

Samantha J. Grayson, Gabriella M. Harari, and Sandra C. Matz. "The impact of stress on personality state expressions". In: *Communications Psychology* 4.1 (Mar. 2026). ISSN: 2731-9121. DOI: 10.1038/s44271-026-00438-3.

Abstract: Who we are at any given moment depends not only on our psychological dispositions but also the situational forces we experience. Here, we explore how stress—an intrapsychic situational force—impacts momentary personality expression. In Study 1 ($N = 792$), we experimentally induced stress, showing that the experience of stress is causally related to lower levels of state Openness, Conscientiousness, Extraversion, Agreeableness and higher levels of state Neuroticism. In Study 2 ($N = 713$; 17,853 observations), we used experience sampling to capture naturally occurring stress and show that stress is associated with lower levels of state Extraversion and Agreeableness, and higher levels of state Neuroticism and Openness in everyday situations. Notably, across both studies, the associations between stress and personality state expressions are largely robust when accounting for momentary affect. Together, our work highlights stress as a distinct and dynamic driver of personality state variability in everyday life.

Groot et al.: Checking the inventory: Illustrating different methods for individual participant data meta-analytic structural equation modeling **Groot-Kan-Jak-2024**

Lennert J. Groot, Kees?Jan Kan, and Suzanne Jak. "Checking the inventory: Illustrating different methods for individual participant data meta-analytic structural equation modeling". In: *Research Synthesis Methods* 15.6 (Aug. 2024), pp. 872–895. ISSN: 1759-2887. DOI: 10.1002/jrsm.1735.

Abstract: Researchers may have at their disposal the raw data of the studies they wish to meta-analyze. The goal of this study is to identify, illustrate, and compare a range of possible analysis options for researchers to whom raw data are available, wanting to fit a structural equation model

(SEM) to these data. This study illustrates techniques that directly analyze the raw data, such as multilevel and multigroup SEM, and techniques based on summary statistics, such as correlation-based meta-analytical structural equation modeling (MASEM), discussing differences in procedures, capabilities, and outcomes. This is done by analyzing a previously published collection of datasets using open source software. A path model reflecting the theory of planned behavior is fitted to these datasets using different techniques involving SEM. Apart from differences in handling of missing data, the ability to include study-level moderators, and conceptualization of heterogeneity, results show differences in parameter estimates and standard errors across methods. Further research is needed to properly formulate guidelines for applied researchers looking to conduct individual participant data MASEM.

Gunn et al.: Characterising patterns of alcohol use among heavy drinkers: A cluster analysis utilising alcohol biosensor data **Gunn-Steingrimsson-Merrill-etal-2021**

Rachel L. Gunn et al. "Characterising patterns of alcohol use among heavy drinkers: A cluster analysis utilising alcohol biosensor data". In: *Drug and Alcohol Review* 40.7 (May 2021), pp. 1155–1164. ISSN: 1465-3362. DOI: 10.1111/dar.13306.

Abstract: Introduction: Previous research has predominately relied on person-level or single characteristics of drinking episodes to characterise patterns of drinking that may confer risk. This research often relies on self-report measures. Advancements in wearable alcohol biosensors provide a multi-faceted objective measure of drinking. The current study aimed to characterise drinking episodes using data derived from a wearable alcohol biosensor. Methods: Participants ($n = 45$) were adult heavy drinkers who wore the Secure Continuous Remote Alcohol Monitoring (SCRAM) bracelet and reported on their drinking behaviours. Cluster analysis was used to evaluate unique combinations of alcohol episode characteristics. Associations between clusters and self-reported person and event-level factors were also examined in univariable and multivariable models. Results: Results suggested three unique clusters: Cluster 1 (most common, slowest rate of rise to and decline from peak), Cluster 2 (highest peak transdermal alcohol concentration and area under the curve) and Cluster 3 (fastest rate of decline from peak). Univariable analyses distinguished Cluster 1 as having fewer self-reported drinks and fewer episodes that occurred on weekends relative to Cluster 2. The effect for number of drinks remained in multivariable analyses. Discussion and Conclusions: This is the first study to characterise drinking patterns at the event-level using objective data. Results suggest that it is possible to distinguish drinking episodes based on several characteristics derived from wearable alcohol biosensors. This examination lays the groundwork for future studies to characterise patterns of drinking and their association with consequences of drinking behaviour.

Hamaker: Analysis of intensive longitudinal data: Putting psychological processes in perspective **Hamaker-2025**

Ellen L. Hamaker. "Analysis of intensive longitudinal data: Putting psychological processes in perspective". In: *Annual Review of Clinical Psychology* 21.1 (May 2025), pp. 379–405. ISSN: 1548-5951. DOI: 10.1146/annurev-clinpsy-081423-022947.

Abstract: Research based on intensive longitudinal data (ILD)—consisting of many repeated measures from one or multiple individuals—is rapidly gaining popularity in psychological science. To appreciate the unique potential of ILD research for clinical psychology, this review begins by examining how our three traditional research approaches fall short when the goal is to investigate processes. It then explores how the analysis of ILD can be used to study a process as it unfolds within a specific person over time but also to study average process features or individual differences therein. By emphasizing the alignment between research questions, data collection, and analytical strategies, the potential of ILD research is further highlighted. It is argued that for future progress it is essential to integrate machine learning and causal inference methods with statistical techniques for ILD and to become more explicit about timescales, time frames, and dynamics in psychological theories.

Hamaker et al.: The fixed versus random effects debate and how it relates to centering in multilevel modeling Hamaker-Muthen-2020

Ellen L. Hamaker and Bengt Muthén. "The fixed versus random effects debate and how it relates to centering in multilevel modeling". In: *Psychological Methods* 25.3 (June 2020), pp. 365–379. ISSN: 1082-989X. DOI: 10.1037/met0000239.

Abstract: In many disciplines researchers use longitudinal panel data to investigate the potentially causal relationship between 2 variables. However, the conventions and concerns vary widely across disciplines. Here we focus on 2 concerns, that is: (a) the concern about random effects versus fixed effects, which is central in the (micro)econometrics/sociology literature; and (b) the concern about grand mean versus group (or person) mean centering, which is central in the multilevel literature associated with disciplines like psychology and educational sciences. We show that these 2 concerns are actually addressing the same underlying issue. We discuss diverse modeling methods based on either multilevel regression modeling with the data in long format, or structural equation modeling with the data in wide format, and compare these approaches with simulated data. We extend the multilevel model with random slopes and discuss the consequences of this. Subsequently, we provide guidelines on how to choose between the diverse modeling options. We illustrate the use of these guidelines with an empirical example based on intensive longitudinal data, in which we consider both a time-varying and a time-invariant covariate.

Haslbeck et al.: Recovering within-person dynamics from psychological time series Haslbeck-Ryan-2022

Jonas M. B. Haslbeck and Oisin Ryan. "Recovering within-person dynamics from psychological time series". In: *Multivariate Behavioral Research* 57.5 (2022), pp. 735–766. ISSN: 1532-7906. DOI: 10.1080/00273171.2021.1896353.

Abstract: Idiographic modeling is rapidly gaining popularity, promising to tap into the within-person dynamics underlying psychological phenomena. To gain theoretical understanding of these

dynamics, we need to make inferences from time series models about the underlying system. Such inferences are subject to two challenges: first, time series models will arguably always be misspecified, meaning it is unclear how to make inferences to the underlying system; and second, the sampling frequency must be sufficient to capture the dynamics of interest. We discuss both problems with the following approach: we specify a toy model for emotion dynamics as the true system, generate time series data from it, and then try to recover that system with the most popular time series analysis tools. We show that making straightforward inferences from time series models about an underlying system is difficult. We also show that if the sampling frequency is insufficient, the dynamics of interest cannot be recovered. However, we also show that global characteristics of the system can be recovered reliably. We conclude by discussing the consequences of our findings for idiographic modeling and suggest a modeling methodology that goes beyond fitting time series models alone and puts formal theories at the center of theory development.

Hecht et al.: A computationally more efficient Bayesian approach for estimating continuous-time models **Hecht-Zitzmann-2020a**

Martin Hecht and Steffen Zitzmann. "A computationally more efficient Bayesian approach for estimating continuous-time models". In: *Structural Equation Modeling: A Multidisciplinary Journal* 27.6 (Mar. 2020), pp. 829–840. ISSN: 1532-8007. DOI: 10.1080/10705511.2020.1719107.

Abstract: Continuous-time modeling is gaining in popularity as more and more intensive longitudinal data need to be analyzed. Current Bayesian software implementations of continuous-time models suffer from rather high, inadequate run times. Therefore, we apply a model reformulation approach to reduce run time. In a simulation study, we investigate the estimation quality and run time gain. We then illustrate our optimized Bayesian continuous-time model estimation and compare it to established continuous-time modeling software using an empirical example. Parameter estimates and inference statistics were very comparable, while run times were very different. Our approach reduces the run times for Bayesian estimations of continuous-time models from hours to minutes.

Hecht et al.: Sample size recommendations for continuous-time models: Compensating shorter time series with larger numbers of persons and vice versa

Hecht-Zitzmann-2020b

Martin Hecht and Steffen Zitzmann. "Sample size recommendations for continuous-time models: Compensating shorter time series with larger numbers of persons and vice versa". In: *Structural Equation Modeling: A Multidisciplinary Journal* 28.2 (July 2020), pp. 229–236. ISSN: 1532-8007. DOI: 10.1080/10705511.2020.1779069.

Abstract: Autoregressive modeling has traditionally been concerned with time-series data from one unit (N = 1). For short time series (T ¡ 50), estimation performance problems are well studied and documented. Fortunately, in psychological and social science research, besides T, another source of information is often available for model estimation, that is, the persons (N ¿ 1). In this work, we illustrate the N/T compensation effect: With an increasing number of persons N at constant T, the model estimation performance increases, and vice versa, with an increasing number of time points T at constant N, the performance increases as well. Based on these observations, we develop sample size recommendations in the form of easily accessible N/T heatmaps for two popular autoregressive continuous-time models.

Hecht et al.: Exploring the unfolding of dynamic effects with continuous-time models: Recommendations concerning statistical power to detect peak cross-lagged effects

Hecht-Zitzmann-2021

Martin Hecht and Steffen Zitzmann. "Exploring the unfolding of dynamic effects with continuous-time models: Recommendations concerning statistical power to detect peak cross-lagged effects". In: *Structural Equation Modeling: A Multidisciplinary Journal* 28.6 (May 2021), pp. 894–902. ISSN: 1532-8007. DOI: 10.1080/10705511.2021.1914627.

Abstract: Cross-lagged panel models have been commonly applied to investigate the dynamic interplay of variables. In such discrete-time models, the size of the cross-lagged effects depends on the length of the time interval between the measurement occasions. Continuous-time modeling allows to explore this interval dependence of cross-lagged effects and thus to identify the maximal "peak" cross-lagged effects. To detect these peak effects, sufficient statistical power is needed. Based on results from a simulation study, we employed machine learning algorithms to identify a highly accurate prediction model. Results are incorporated into a Shiny App (available at https://psychtools.shinyapps.io/ ContinuousTimePowerCalculation) for easy power calculations. Although limitations apply, our results might be helpful for study planning.

Hisler et al.: Neuroticism as the intensity, reactivity, and variability in day-to-day affect **Hisler-Krizan-DeHart-etal-2020**

Garrett C. Hisler et al. "Neuroticism as the intensity, reactivity, and variability in day-to-day affect". In: *Journal of Research in Personality* 87 (Aug. 2020), p. 103964. ISSN: 0092-6566. DOI: 10.1016/j.jrp.2020.103964.

Abstract: Neuroticism has been linked to typical levels of affect, affect reactivity to negative events, and variability in affect over time. However, the intercorrelations among these characteristics make it unclear whether neuroticism reflects unique variance in each of these aspects of emotional life. Data from two daily-diary samples revealed that neuroticism was associated with average levels and variability of positive and negative affect and reactivity of negative affect to stressors, but was only uniquely related to mean levels of positive and negative affect. Findings highlight the substantial overlap in affect indices, suggesting that mean levels of affect, at the very least, are at the core of neuroticism, and reveal the need for further research using more nuanced approaches.

Hoekstra et al.: Heterogeneity in individual network analysis: Reality or illusion? **Hoekstra-Epskamp-Borsboom-2023**

Ria H. A. Hoekstra, Sacha Epskamp, and Denny Borsboom. "Heterogeneity in individual network analysis: Reality or illusion?" In: *Multivariate Behavioral Research* 58.4 (2023), pp. 762–786. ISSN: 1532-7906. DOI: 10.1080/00273171.2022.2128020.

Abstract: The use of idiographic research techniques has gained popularity within psychological research and network analysis in particular. Idiographic research has been proposed as a promising avenue for future research, with differences between idiographic results highlighting evidence for radical heterogeneity. However, in the quest to address the individual in psychology, some classic statistical problems, such as those arising from sampling variation and power limitations, should not be overlooked. This article aims to determine to what extent current tools to compare idiographic networks are suited to disentangle true from illusory heterogeneity in the presence of sampling error. To this end, we investigate the performance of tools to inspect heterogeneity (visual inspection, comparison of centrality measures, investigating standard deviations of random effects, and GIMME) through simulations. Results show that power limitations hamper the validity of conclusions regarding heterogeneity and that the power required to assess heterogeneity adequately is often not realized in current research practice. Of the tools investigated, inspecting standard deviations of random effects and GIMME proved the most suited. However, all tools evaluated leave the door wide open to misinterpret all observed variability in terms of individual differences. Hence, the current paper calls for caution in the use and interpretation of new time-series techniques when it comes to heterogeneity.

Hunter: State space mixture modeling: Finding people with similar patterns of change

Hunter-2024

Michael D. Hunter. "State space mixture modeling: Finding people with similar patterns of change". In: *Multivariate Behavioral Research* 59.6 (2024), pp. 1253–1269. ISSN: 1532-7906. DOI: 10.1080/00273171.2023.2261224.

Abstract: Increasingly, behavioral scientists encounter data where several individuals were measured on multiple variables over numerous occasions. Many current methods combine these data,

assuming all individuals are randomly equivalent. An extreme alternative assumes no one is randomly equivalent. We propose state space mixture modeling as one possible compromise. State space mixture modeling assumes that unknown groups of people exist who share the same parameters of a state space model, and simultaneously estimates both the state space parameters and group membership. The goal is to find people that are undergoing similar change processes over time. The present work demonstrates state space mixture modeling on a simulated data set, and summarizes the results from a large simulation study. The illustration shows how the analysis is conducted, whereas the simulation provides evidence of its general validity and applicability. In the simulation study, sample size had the greatest influence on parameter estimation and the dimension of the change process had the greatest impact on correctly grouping people together, likely due to the distinctiveness of their patterns of change. State space mixture modeling offers one of the best-performing methods for simultaneously drawing conclusions about individual change processes while also analyzing multiple people.

Hunter et al.: What ergodicity means for you **Hunter-Fisher-Geier-2024**

Michael D. Hunter, Zachary F. Fisher, and Charles F. Geier. "What ergodicity means for you". In: *Developmental Cognitive Neuroscience* 68 (Aug. 2024), p. 101406. ISSN: 1878-9293. DOI: 10.1016/j.dcn.2024.101406.

Abstract: This paper explores the relation between within-person and between-person research designs using the concept of ergodicity from statistical mechanics in physics. We demonstrate the consequences of ergodicity using several real data examples from previously published studies. We then create several simulated examples that illustrate the independence of within-person processes from between-person differences, and pair these examples with analytic results that reinforce our conclusions. Finally, we discuss the plausibility of ergodicity being the general rule rather than the exception for social and behavioral processes, address common arguments against heeding the implications of ergodicity for behavioral research, and offer several possible solutions.